

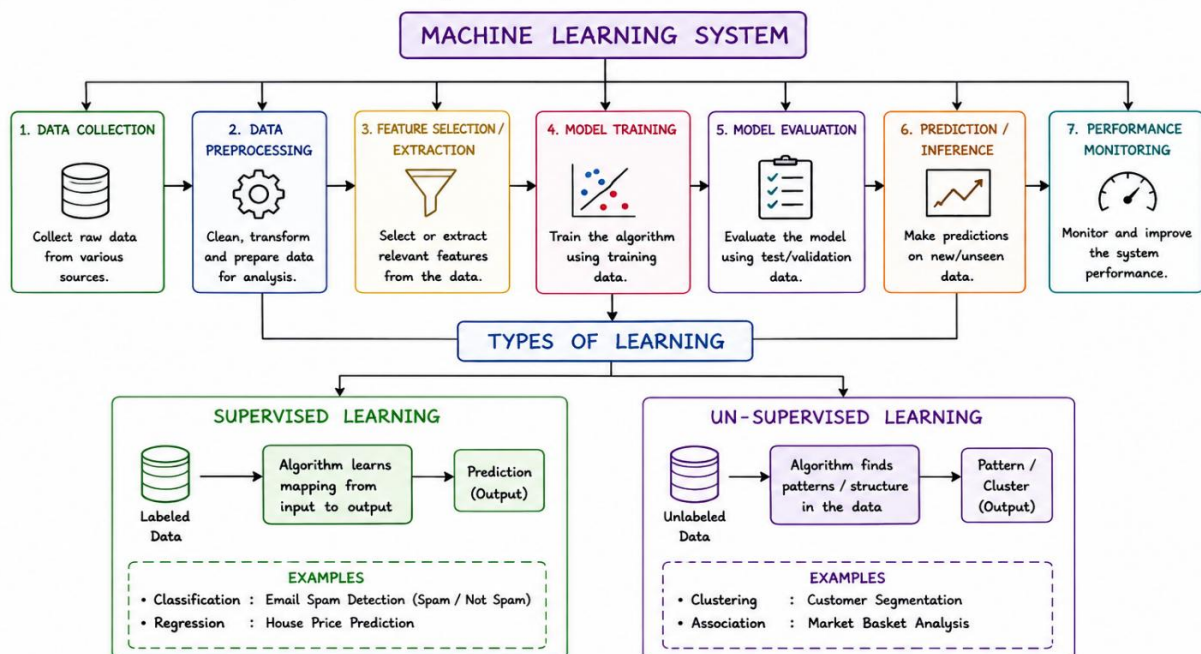
SIMATS ENGINEERING

ASSESSMENT TOOL 2

CONCEPT MAPPING

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



Explanation:

- A Machine Learning (ML) system learns patterns from data to make accurate predictions or decisions without being explicitly programmed.
- The process starts with **data collection**, followed by **data preprocessing**, where missing values, noise, and inconsistencies are removed.
- **Feature engineering** is performed to select or create important features that improve the model's performance.
- The processed data is used for **model training** using suitable algorithms such as Decision Tree, Logistic Regression, or Random Forest.
- The trained model is evaluated using metrics like **Accuracy, Precision, Recall, and F1-score** to measure its performance.
- Finally, the validated model is **deployed** to make predictions on new data in real-world applications such as healthcare, finance, and spam detection.

Learning Types:

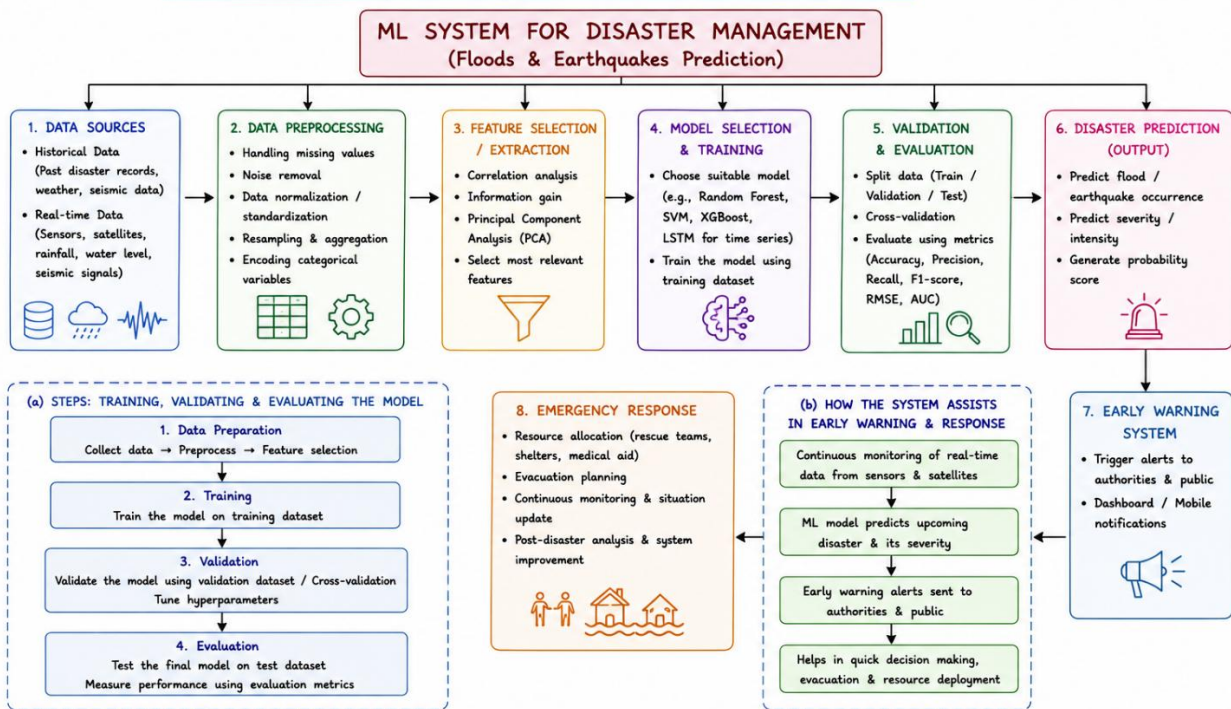
- **Supervised Learning:** Uses labeled data for classification and regression tasks.
- **Unsupervised Learning:** Uses unlabeled data to identify hidden patterns or clusters.
- **Reinforcement Learning:** Learns by interacting with the environment and receiving rewards or penalties.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response..

- Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data.



Explanation:

- A Machine Learning-based Disaster Management System predicts natural disasters such as floods, earthquakes, cyclones, and landslides using historical and real-time data.
- The system begins with **data collection** from sensors, satellites, weather stations, and historical disaster records.
- During **data preprocessing**, missing values, noise, and duplicate records are removed to improve data quality.
- Feature selection** identifies important parameters such as rainfall, temperature, humidity, wind speed, and seismic activity for accurate prediction.
- The processed data is used to **train machine learning models** such as Decision Tree, Random Forest, or Neural Networks to predict disaster occurrence.

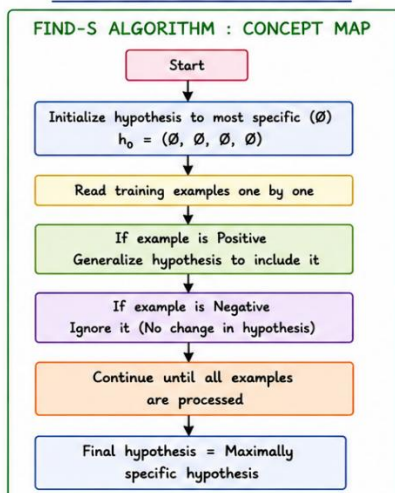
- The trained model is evaluated for accuracy and deployed to provide **early warnings**, helping authorities take timely preventive actions and reduce loss of life and property.

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

3. FIND-S ALGORITHM (Play Tennis)



Dataset :

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

STEP-BY-STEP FIND-S ALGORITHM (Consider Positive Examples Only)

Step	Training Example (Only Positive)	Example Details (Outlook, Temp, Humidity, Wind)	Hypothesis (h) After Update	Explanation
1	E ₃	(Overcast, Hot, High, Weak)	(Overcast, Hot, High, Weak)	Initialize to first positive example
2	E ₄	(Rainy, Mild, High, Weak)	(?, ?, High, Weak)	Generalize differing attributes
3	E ₅	(Rainy, Cool, Normal, Weak)	(?, ?, ?, Weak)	Generalize further
4	E ₇	(Overcast, Mild, Normal, Strong)	(?, ?, ?, ?)	Generalize all differing attributes

FINAL MAXIMALLY SPECIFIC HYPOTHESIS : $h = (?, ?, ?, ?)$

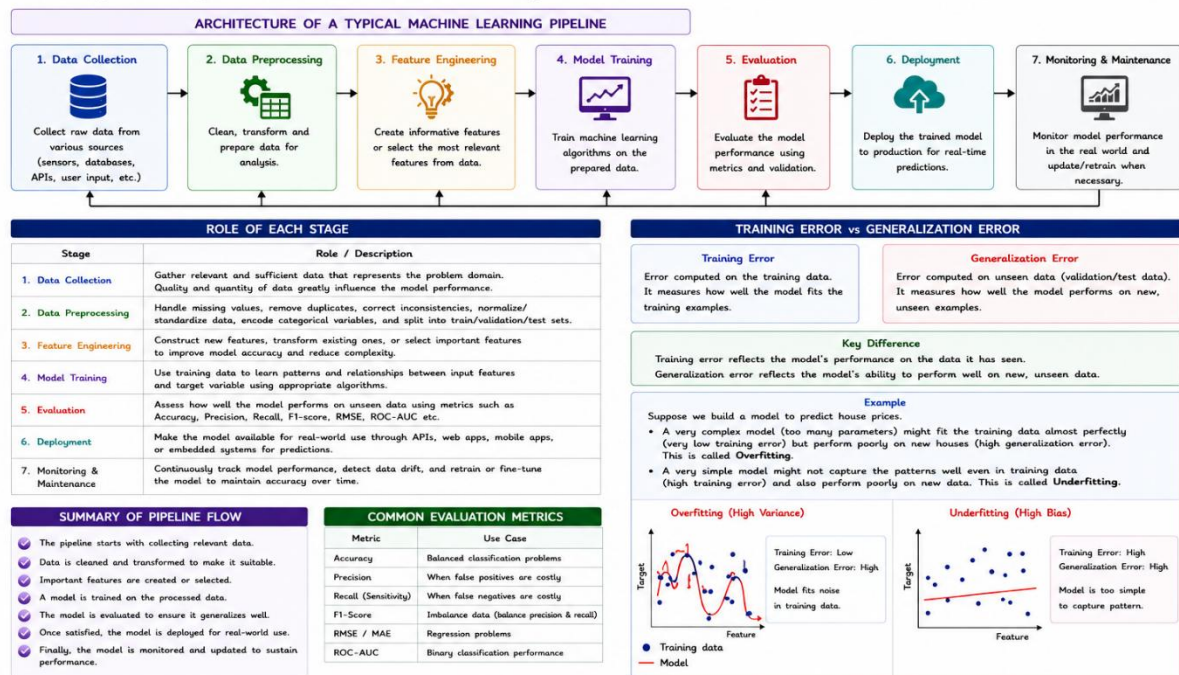
Note : All negative examples (E₁, E₂, E₆, E₈) are ignored.

Explanation:

- The **Find-S algorithm** is a supervised learning algorithm that finds the **most specific hypothesis** consistent with all positive training examples.
- It starts with the **most specific hypothesis** and considers only the **positive examples**, ignoring all negative examples.
- For each positive example, the hypothesis is updated by replacing differing attribute values with a general symbol (?) while keeping matching attributes unchanged.
- The algorithm continues this process until all positive examples are processed, resulting in the final hypothesis.
- The final hypothesis represents the most specific rule that correctly classifies all positive training examples and can be used to predict new instances.
- Find-S is simple and efficient but **cannot handle inconsistent data or negative examples**, making it suitable only for datasets with reliable positive examples.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.



Explanation:

- A **Machine Learning pipeline** is a sequence of steps used to build, evaluate, and deploy a machine learning model efficiently.
- The pipeline begins with **data collection**, followed by **data preprocessing**, where missing values, duplicate records, and noise are removed to improve data quality.
- Feature engineering** is then performed to select or create the most relevant features that improve model performance.
- The processed data is used for **model training** using algorithms such as Decision Tree, Random Forest, or Logistic Regression. The model is then evaluated using metrics like **Accuracy, Precision, Recall, and F1-score** before deployment.
- Training error** is the error made by the model on the training dataset, while **generalization error** is the error on new, unseen test data. A model with low training error but high generalization error indicates **overfitting**.
- Finally, the validated model is **deployed** to make real-time predictions in applications such as healthcare, finance, spam detection, and recommendation systems.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.

(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).

(b) Explain feature scaling and handling missing data.

(c) Describe how overfitting can be detected and avoided.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

Problem Statement
Design an ML system to predict house prices (continuous target) using features: Location, Size (sq.ft), Number of Rooms, and Amenities.

System Overview

(a) Choice of Regression Model and Justification
Chosen Model: **Random Forest Regressor**
Justification:

- House price prediction is a complex non-linear problem with interactions between features.
- Random Forest is an ensemble of decision trees that can capture non-linear relationships without requiring feature scaling.
- It handles both numerical and categorical features well (after encoding).
- Robust to outliers and noise and has low variance compared to a single decision tree.
- Provides feature importance, which helps in understanding key drivers of price.
- Generally gives higher accuracy than simple Linear Regression and more stable than a single Decision Tree.

Model	Pros	Cons	When Suitable
Linear Regression	Simple, interpretable, fast	Assumes linearity, sensitive to outliers	When relationship is approximately linear
Decision Tree Regressor	Captures non-linear patterns, easy to interpret	High variance, prone to overfitting	Small datasets, when interpretability is critical
Random Forest Regressor	High accuracy, handles non-linearity, robust, feature importance	Less interpretable than single tree, more computation	Most real-world house price prediction problems

(b) Feature Scaling and Handling Missing Data
Feature Scaling

- Features like Size (in sq.ft) and Number of Rooms may have larger ranges than others.
- Scaling ensures all features contribute equally to the model and improves performance (important for Linear Regression and other distance-based models).

Common Methods

- Standardization (Z-score scaling): $z = \frac{x - \mu}{\sigma}$
- Min-Max Scaling: $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$

- Tree-based models (e.g., Random Forest) are less sensitive to scaling, but scaling is still good practice.

Handling Missing Data

- Missing values can degrade model performance.

Strategies:

- Numerical Features:
 - Impute with mean/median (simple and effective)
 - Or use KNN/Regression Imputation
- Categorical Features:
 - Impute with mode (most frequent)
- If missingness is significant, add a missing indicator feature (1 if missing, else 0).
- Remove rows/columns only if missing data is very high (> 40-50%).

Summary
We use a Random Forest Regressor for accurate house price prediction because it handles non-linear relationships, is robust, and provides high performance. We preprocess data by scaling features and handling missing values through imputation. Overfitting is detected using validation metrics and learning curves and avoided through regularization, tuning, pruning, and cross-validation.

Example Dataset (Sample)

Location (Encoded)	Size (sq.ft)	No. of Rooms	Amenities (Count)	Age of House (years)	Price (₹ Lakh)
0.35	1200	2	5	10	45.0
0.80	1800	3	8	5	78.0
0.25	1000	2	3	15	38.0
0.60	2400	4	10	2	105.0
0.10	800	1	2	20	25.0
...	...	...	...	...	...

(c) Detecting and Avoiding Overfitting

How to Detect Overfitting

Overfitting occurs when the model learns noise in training data and performs poorly on unseen data.

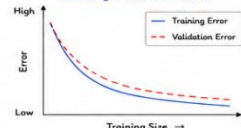
Detection methods:

- Train-Test Split: Large gap between training score and test score indicates overfitting.
- Cross-Validation: High variance across folds indicates overfitting.
- Learning Curve: Training error keeps decreasing while validation error starts increasing.
- Evaluation Metrics on unseen test set (e.g., RMSE, MAE, R^2).

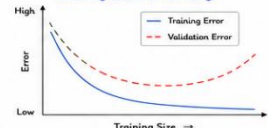
How to Avoid Overfitting

- Use sufficient training data.
- Feature Selection: Remove irrelevant/redundant features.
- Regularization (if using Linear Regression):
 - Ridge (L2), Lasso (L1)
- Hyperparameter Tuning (for Random Forest):
 - Limit max_depth
 - Set min_samples_split
 - Set min_samples_leaf
 - Limit n_estimators (avoid too high)
- Use Cross-Validation for model selection.
- Pruning (for Decision Trees).

Learning Curve (Good Fit)



Learning Curve (Overfitting)



Model Evaluation Metrics (Regression)

Metric	Formula	Description
MAE	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Average absolute difference between actual and predicted price.
MSE	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	Average squared difference (penalizes large errors more).
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Square root of MSE; in same units as price.
R^2 Score	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Proportion of variance explained by the model (closer to 1 is better).

Explanation:

- A **House Price Prediction System** uses machine learning to estimate the price of a house based on features such as **location, size, number of rooms, and amenities**.
- The collected housing data is **preprocessed** by handling missing values, removing duplicates, and applying **feature scaling** to improve model performance.

- A suitable regression model such as **Random Forest Regression** is selected because it provides high accuracy, handles nonlinear relationships, and reduces overfitting compared to a single Decision Tree.
- The model is trained using the processed dataset and predicts the price of new houses based on the input features.
- **Overfitting** is detected by comparing training and testing performance. It can be reduced using **cross-validation, pruning, regularization, and by increasing the training data.**
- The final trained model helps buyers, sellers, and real estate agencies estimate accurate house prices for informed decision-making.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

- (a) Describe the working of the K-Means algorithm with a step-by-step example.**
- (b) Explain how the value of K is chosen using the Elbow Method.**
- (c) Compare K-Means with Hierarchical Clustering.**

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

Clustering (Unsupervised Learning)

Clustering is a technique in unsupervised learning that groups similar data points together based on their inherent patterns or similarity, without using any class labels. The goal is to maximize similarity within a cluster and minimize similarity between clusters.

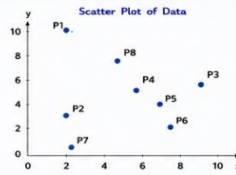
(a) K-Means Algorithm with Step-by-Step Example

K-Means Algorithm (Working)

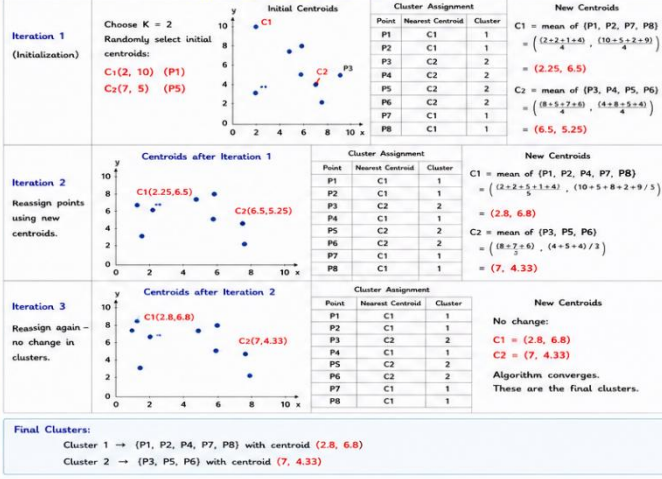
- Step 1: Choose the number of clusters K.
- Step 2: Randomly initialize K centroids.
- Step 3: Assign each data point to the nearest centroid (based on Euclidean distance).
- Step 4: Recalculate the centroid of each cluster (mean of all points assigned to that cluster).
- Step 5: Repeat Steps 3 and 4 until centroids do not change (convergence).

Example Dataset (2-D Points)

Point	x	y
P1	2	10
P2	2	5
P3	8	4
P4	5	8
P5	7	5
P6	6	4
P7	4	9



Step-by-Step Execution (Let K = 2)

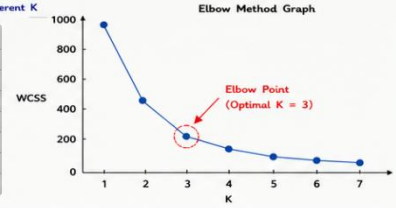


(b) Choosing the Value of K using the Elbow Method

Elbow Method: Run K-Means for different values of K and compute Within-Cluster Sum of Squares (WCSS). Choose K at the "elbow" point where the rate of decrease sharply changes.

Example: WCSS for different K

K	WCSS
1	980
2	452
3	235
4	143
5	98
6	72
7	61



After K = 3, the reduction in WCSS is small → hence K = 3 is optimal.

(c) Comparison: K-Means vs Hierarchical Clustering

Aspect	K-Means Clustering	Hierarchical Clustering
Type	Partition-based	Hierarchy-based
Number of clusters	Must be specified (K)	Not required in advance
Output	Single flat clustering	Dendrogram (multi-level clusters)
Working	Start with K centroids, assign points & update iteratively	Builds a tree of clusters either bottom-up (agglomerative) or top-down (divisive)
Distance metric	Usually Euclidean distance	Any distance/similarity metric
Scalability	Efficient for large datasets (O(nkt))	Computationally expensive (O(n ²) to O(n ³))
Cluster shape	Works well for spherical, compact clusters	Can capture any shape clusters
Handling noise/outliers	Sensitive to outliers	More robust (with suitable linkage)
Flexibility	Less flexible	More flexible
Interpretability	Easy to interpret	Dendrogram gives more insight
Examples	Customer segmentation, image compression	Gene clustering, document clustering, taxonomy

Key Takeaway

- Clustering groups similar data without labels.
- K-Means is simple, fast, and effective for large datasets with compact clusters.
- The Elbow Method helps choose an appropriate number of clusters K.
- Hierarchical clustering provides a hierarchy of clusters and works well for small datasets and complex structures.

Explanation:

- Clustering** is an **unsupervised learning** technique used to group similar data points into clusters without using labeled data.
- The **K-Means algorithm** begins by selecting the number of clusters (**K**), initializing centroids, assigning each data point to the nearest centroid, and updating the centroids repeatedly until the clusters become stable.
- The **Elbow Method** is used to determine the optimal value of **K** by plotting the Within-Cluster Sum of Squares (WCSS). The point where the graph forms an "elbow" is chosen as the best value of **K**.
- K-Means Clustering** is simple, fast, and suitable for large datasets, but it requires the value of **K** to be specified in advance and is sensitive to outliers.
- Hierarchical Clustering** builds a tree-like structure (dendrogram) of clusters and does not require a predefined value of **K**, but it is computationally more expensive than K-Means.
- Clustering is widely used in applications such as **customer segmentation**, **image segmentation**, **document classification**, **recommendation systems**, and **medical data analysis**.

